

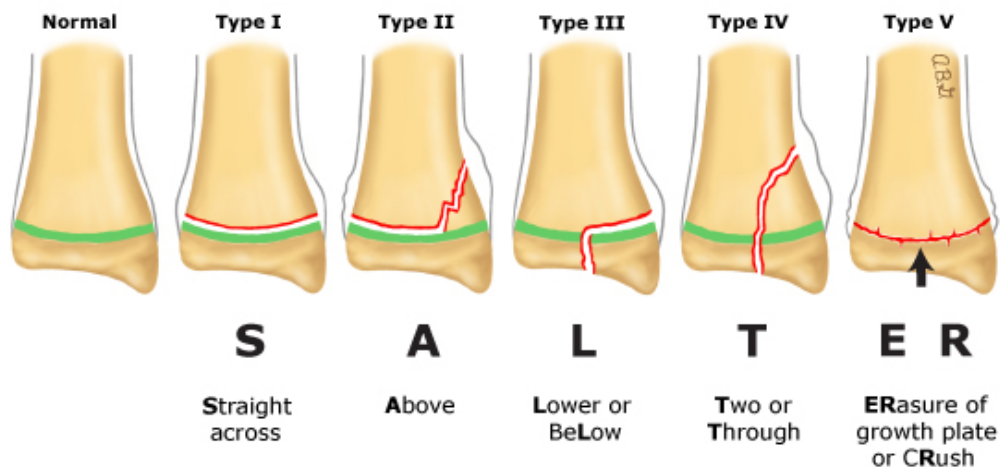
Clinical Practice Guidelines: Upper limb injuries presenting to Children's Emergency

When to order Xrays

- Signs and symptoms of bony injury or deformity or concerns of foreign body
- Suspected OM/septic arthritis - though Xray can be normal in early stages of infections
- Suspicion of non-accidental injury
- Post-reduction of displaced fracture or dislocation except for reduction of pulled elbow

Salter Harris classification for Physeal injuries

- I: Fracture occurs through the physes; subtle widening of physes with no obvious fracture line
- II: Fracture involves metaphyses (above).
- III: Fracture involves epiphysis (below).
- IV: Comprising II + III (through and through)
- V: Compression force



(Diagram obtained from Uptodate)

Relevance of Salter Harris classification and Mx of physeal injuries:

Most of type I and II injuries are treated by closed reduction and cast immobilisation.

Most of type III and IV injuries require ORIF.

Most heal quickly and recover fully, with a minority with growth arrest which can result in deformity and impaired function.

Description of fractures

- Open / close
- Displaced / undisplaced / angulated
- Transverse / oblique / spiral / comminuted
- Complete / incomplete
- Site – shaft / distal / proximal / intra-articular

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Optimal placement of the forearm cast

- Fracture at proximal 1/3 forearm – patient's palm in supine (facing up)
- Fracture at middle 1/3 forearm – patient's arm in lateral position
- Fracture at distal 1/3 forearm – patient's palm in prone (facing down)
- Displaced / angulated # - place on side of displacement of the **distal fragment**. Eg: dorsal displacement = dorsal backslab (see diagram below).



This is a distal ulnar shaft fracture with dorsal displacement of the distal fragment. Therefore the backslab should be placed dorsally with the wrist in slight flexion.



Testing for nerve function

- Ulnar nerve (cross middle finger over index finger)
- Radial nerve (thumbs up, wrist extension, MCPJ extension)
- Median ("ok" sign)

Indications for referral to orthopaedics team

- All open fractures (these patients need tetanus, IV antibiotics, NBM, wound exploration in OT)
- All pathological fractures
- Fractures associated with neurovascular compromise (weakened pulse, nerve palsy, pale/cold limb)
- All fracture dislocations
- All suspected septic arthritis or osteomyelitis
- Specifics of each upper limb fracture covered below for referral while in CE

Disposition

With all fractures, please discharge with analgesia and cast care advice.
TCU will be to paed's orthopaedics for all fractures; fractures of the hand distal to distal radio-ulnar joint (DRUJ) will be followed up with HRM instead.

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UPPER LIMB INJURIES

Anterior dislocation of shoulder

- Views to order - AP/Lat view and axillary view. A modified axillary view (velpeau view) can be considered for patients who are unable to abduct their arm.
- The presence of Bankart or Hill-Sachs lesions, which do not need immediate repair, can result in recurrent shoulder dislocations.
- Test axillary nerve function (deltoid muscle function and regimental badge sensation).
- Mx:

Reduce immediately under sedation (intranasal fentanyl with nitrous / ketamine / propofol)
Repeat XRay post reduction
Immobilise with arm sling (collar and cuff if concomitant # of the proximal humerus)
TCU ortho in 1 week

Clavicle #

- Look for the presence of pneumothorax.
- Mx:

Arm sling, no contact sports for 6 weeks, TCU ortho in 1 week
Refer ortho if

- o Tenting of skin
- o Displaced medial clavicular #
- o Mid-shaft clavicular # with >2cm shortening
- o Lateral clavicular # with significant medial clavicular displacement
- o No bony contact or severely comminuted fractures

- Discharge advice- inform parents that lump will develop at fracture site and may be visible for up to 1 year. Give adequate analgesia as pain can be significant in the first few days post injury.

Proximal humeral # / Humeral neck #

- Most are SH I or II injuries.
- Test axillary nerve function (deltoid muscle function and regimental badge sensation).
- Most do not require M+R due to great remodelling potential. No role for M+R in ED.
- Mx:

Collar and cuff, TCU ortho 1/52
Refer ortho:

- > 60 deg angulation and >50% displacement in <12 y.o.
- > 30 deg angulation and >30% displacement in >12 y.o.

Humeral shaft #

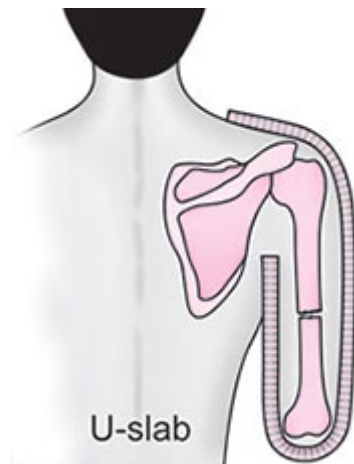
- Does not need M+R due to risk of radial nerve injury. Alignment is achieved by gravity.
- Consider NAI for spiral humeral # as these are typically caused by indirect twisting from a fall or intentionally.
- Watch for radial nerve palsy which can occur with a # occurring at junction of middle and distal 1/3 of shaft.
- Mx:

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Proximal third #: Collar and cuff, TCU ortho 1/52
Middle third and distal third #: U-slab (see below), TCU ortho 1/52
Refer ortho:

- >20 degree varus/valgus or >20 degree procurvatum
- >15 degree rotation malalignment
- >2cm shortening

-Discharge advice: sleep in a propped up position as gravity helps to reduce #, regular analgesia, return to CE if pain worsens or sensation changes suddenly



U slab:

Starts at the axilla, continues along the medial side of the arm, wraps around the elbow, continues along the lateral aspect of the arm and ends at the shoulder. The two limbs of the slab are separated, therefore it is called a "U slab".

Medial epicondylar #

- Look specifically for ulnar nerve palsy.
- Half of the cases occur with elbow dislocation – look for the medial epicondyle (which may be displaced) in cases of elbow dislocation.
- Need to distinguish medial epicondylar # from medial condyle #. Medial condylar fracture is rare, but this is an intra-articular fracture extending into elbow joint that needs ORIF
- Mx:

Undisplaced or minimally displaced (<5mm): Above elbow backslab with elbow in 90 degrees, TCU ortho in 1 week

Displaced (5mm or more): Refer ortho

Lateral condylar #

- Risk of valgus deformity.
- Note that part of the lateral condyle is cartilaginous and may not be seen on XR, which means the actual fracture may be more prominent than what is seen.
- This is where the forearm extensors originate from thus making it an unstable injury, prone to malunion, and therefore has a lower threshold for surgical fixation.
- Mx:

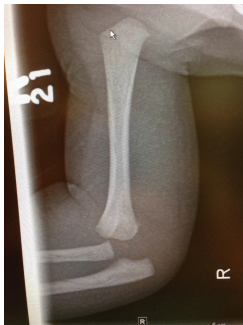
Undisplaced or minimally displaced (<2mm): Above elbow backslab with elbow in 90 degrees, TCU ortho early (in 3 days) due to potential for displacement and need for surgery.

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Displaced (>2mm)- Refer ortho. Will require surgical management.

Physeal separation of distal humerus

- Horizontal fracture through physal plate of distal humerus.
- Rare and easily missed as capitellum ossification typically occurs at 1 year old and most/all of the distal humerus epiphysis cannot be visualised directly on Xray for young children.
- High index of suspicion if patient <2 years old, elbow swelling with appearance of what looks like lateral condyle fracture on Xray.
- Posterior and medial displacement of radius and ulna.
- High chance of cubitus varus deformity if not stably reduced or fixed early surgically.
- Mx: refer ortho



Physeal separation of distal humerus where there is posteromedial displacement of the distal fragment (Images from orthobullets)

Supracondylar

- Extension-type fractures are more common (98%) - distal fragment is displaced posteriorly
- Flexion-type fracture rare - distal fragment displaced anteriorly
- Note that radiographs need to be a true lateral film to be able to comment on whether anterior humeral line (AHL) cuts the capitellum

Gartland classification for extension-type supracondylar fractures (see diagram below):

Type	Finding	Management
I	Undisplaced / fat pad sign**	Above elbow backslab with elbow in flexion
II	Angulated # with intact posterior cortex	<u>a) If AHL* cuts capitellum and no medial comminution#</u> -Mx: Above elbow backslab <u>b) If AHL does not cut capitellum or presence of medial comminution#:</u> -Mx: M+R with satisfactory alignment (ie AHL cutting capitellum), then above elbow backslab <u>c) Rotational component to suggest type IIb</u> -Mx: Refer ortho for surgical intervention
III	Displaced distal fragment with no cortical contact	Refer ortho for surgical intervention High risk of neurovascular compromise

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	Place saturation probe on affected side to ensure good waveform
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Note that above elbow backslab should extend up as close to axilla as possible, and down to MCPJ.

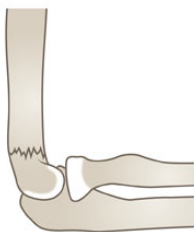
*AHL = Anterior humeral line

#Presence of medial comminution – this refers to medial cortical bone being fragmented. Note that this is not part of Gartland classification.

**Fat Pad Sign

- When a joint effusion is present, it distends the capsule and displaces the fat pads away from the bone.
- A small anterior fat pad is normal but if it is displaced anteriorly (like a sail), it is abnormal
- A posterior fat pad is **always** abnormal- it signifies a large effusion.
- Therefore in the context of trauma, an effusion indicates that a significant injury has occurred even if a fracture is not seen.

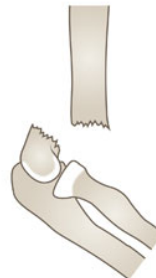
Type I



Type II



Type III



Diagrams obtained from Royal Children's Hospital Clinical Practice Guidelines website (www.rch.org.au)

Olecranon

-Always look for radial head dislocation (Monteggia variant) by drawing the radiocapitellar line

-Mx:

- Undisplaced: Above elbow backslab in 90 degrees elbow flexion, TCU 1/52
- Displaced: Refer ortho

Radius and ulna

-In single bone fractures, be careful to inspect the proximal and distal radioulnar joints clinically and radiologically

a) Monteggia

-Proximal ulnar # with radial head dislocation

-Monteggia variant = radial head dislocation in the presence of ulna plastic deformity

-Look for radial nerve palsy and posterior interosseous nerve palsy

b) Galeazzi # (very uncommon)

-Distal 1/3 radial # + dislocation of DRUJ

-Mx: Refer all Monteggia and Galeazzi # to orthopaedics

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Radial neck / head

-Need to look for radial nerve palsy (wrist drop), posterior interosseous nerve palsy (difficulty with MCPJ extension)

-Need to look for concomitant ulna # - in the presence of radial head dislocation, do a dedicated radius/ulna view to rule out Monteggia fracture.

-Mx:

Undisplaced: Above elbow backslab, TCU 1/52 Displaced (> 30 degrees angulation): Refer ortho Presence of radial head dislocation: Attempt reduction in CE; Refer ortho if unsure.

Radius / ulna shaft

-Mx:

Undisplaced: Sugartong U slab* / Above elbow backslab with elbow in 90 degrees, TCU 1/52 Displaced: Reduce. Accept if no angulation, acceptable if bayoneted. a) For middle and distal shaft # Degree of angulation: in <10 years old, can allow up to 15 degrees angulation in >10 years old, can allow up to 10 degrees angulation b) For proximal third shaft # - to refer orthopaedics



Sugartong U slab: Starts at metacarpals dorsally, runs to the elbow and wraps around the elbow to the volar surface of the forearm, ends at palmar surface of the metacarpals.

Diagram obtained from American Academy of Family Physicians webpage
(<https://www.aafp.org/afp/2009/0901/p491.html>)

Distal radius / ulna

-Mx:

Undisplaced: Below elbow backslab, TCU 1/52 Displaced/Angulated: In <10 years old, can allow up to 15 deg angulation; In >10 years old, can allow up to 10 deg angulation.
NB: For all distal radius/ulna fractures requiring M+R, place an above elbow backslab instead of a short slab to avoid pronation / supination.
Backslab to be placed on site of distal displacement, e.g.: dorsally angulated fracture (distally) would mean a dorsal backslab. Refer to orthopaedics if inability to achieve acceptable reduction.

Hand fractures

-Always palpate anatomical snuff box- tenderness over this area requires a scaphoid view Xray

-Refer to HRM for

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- a) Open fractures
- b) Hand fracture with dislocation
- c) Unstable fractures / failed M+R
- d) Rotational component / loss of hand cascade

-Note that a displaced intra-articular fractures can be given an early HRM clinic (2-3 days). If in doubt, please refer.

-Full volar splint (ie ulnar gutter, thumb spica, intrinsic plus depending on which digit is fractured) for base of proximal phalangeal fracture or more proximal to that; finger splint for all other fractures.

- Ulnar gutter– for fracture of 4th or 5th base of proximal phalanx or metacarpal #
- Intrinsic plus – for fracture of 2nd or 3rd base of proximal phalanx or metacarpal #
- Thumb spica – for fracture of base of proximal phalanx (thumb) or 1st metacarpal #

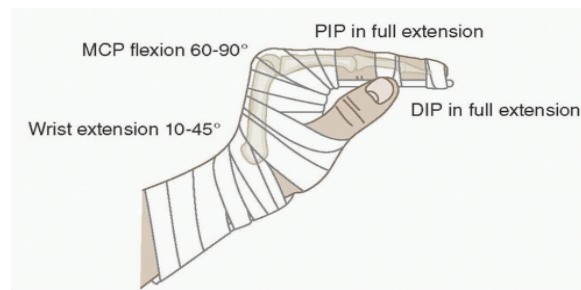
-Place a gauze in between fingers that are splinted together to prevent maceration

-For all splints - to be done when hand is in a SAFE position (see below) –

Wrist 10 to 45 degrees of extension

MCP joint 60 to 90 degrees of flexion.

PIP and DIPJ in full extension



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Author: Dr Ivy Ang. Vetted by: Dr Ian Tan, Dr Mohite Sharad; Dr Sharon Tan, Dr Andrew Lim.

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